

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Computer Science and Engineering

### CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

|                |                                                                                                                                                 |               |                                                         |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------------------------------------------------|
| Course Code:   | CSA10 / CSA1063                                                                                                                                 | Course Title: | Software Engineering                                    |
| CO Assessed:   | CO2                                                                                                                                             | Assessment:   | Assessment Tool 2 – Scenario-Based Requirement Analysis |
| Weightage:     | 20%                                                                                                                                             | Total Marks:  | 25                                                      |
| CO2 Statement: | <i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i> |               |                                                         |
| Student Name:  | ARTHI A                                                                                                                                         | Reg. No.:     | 192525152                                               |
| Date:          | 06.08.2026                                                                                                                                      | Duration:     | 60 minutes                                              |

#### Code of Conduct

I ARTHI A (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: \_\_\_\_\_

Arthi A

### Annexure A – Scenario-Based Requirement Analysis

#### Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

#### Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

- 1. Study and Analyze the Scenario**
  - Carefully understand the given problem statement.
  - Identify the business domain, stakeholders, existing challenges, and system objectives.
- 2. Identify Functional and Non-Functional Requirements**
  - List all functional requirements required by the proposed system.
  - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
- 3. Identify Actors and Use Cases**
  - Identify all primary and secondary actors interacting with the system.
  - List the major use cases associated with each actor.
  - Draw a Use Case Diagram representing the interactions between actors and the system.
- 4. Prepare the Software Requirement Specification (SRS)**

Develop an SRS document containing:

  - Introduction
  - Problem Description

- Scope of the System
  - Functional Requirements
  - Non-Functional Requirements
  - Use Case Descriptions
  - Data Requirements
  - External Interface Requirements
  - System Features
5. **Identify Assumptions and Constraints**
- List the assumptions considered while developing the system.
  - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
6. **Validate Requirement Completeness and Consistency**
- Verify whether all stakeholder requirements have been addressed.
  - Check for ambiguity, redundancy, inconsistency, and missing requirements.
  - Suggest improvements to enhance the quality and completeness of the requirement specification.

#### Rubrics for Calculation

| Criteria                                                        | Excellent (4 Marks)                                                                                                        | Good (3 Marks)                                                     | Satisfactory (2 Marks)                                          | Needs Improvement (1 Mark)                                           |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------|
| <b>Scenario Understanding and Problem Analysis</b>              | Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.   | Demonstrates good understanding with minor omissions.              | Basic understanding with limited analysis.                      | Poor understanding; major aspects missing or incorrect.              |
| <b>Functional and Non-Functional Requirement Identification</b> | Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.                | Identifies most requirements with minor classification errors.     | Identifies only basic requirements; several omissions.          | Requirements are incomplete, inaccurate, or poorly classified.       |
| <b>Actor Identification and Use Case Modeling</b>               | Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards. | Most actors and use cases identified; diagram has minor errors.    | Limited actors/use cases identified; diagram partially correct. | Actors/use cases missing or incorrect; diagram absent or inaccurate. |
| <b>Software Requirement Specification</b>                       | SRS is well-structured, complete, professionally organized, and                                                            | SRS is organized with minor missing sections or formatting issues. | SRS includes only essential sections with limited detail.       | SRS is incomplete, poorly organized, or                              |

|                                                             |                                                                                                                                                               |                                                                      |                                                             |                                                                    |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------|
| <b>(SRS) Documentation</b>                                  | includes all required sections.                                                                                                                               |                                                                      |                                                             | lacks essential components.                                        |
| <b>Assumptions, Constraints, and Requirement Validation</b> | Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification. | Identifies most assumptions and constraints; validation is adequate. | Limited assumptions/constraints; validation is superficial. | Assumptions, constraints, and validation are missing or incorrect. |

| Criteria                                                 | Mark |
|----------------------------------------------------------|------|
| Scenario Understanding and Problem Analysis              | /5   |
| Functional and Non-Functional Requirement Identification | /5   |
| Actor Identification and Use Case Modeling               | /5   |
| Software Requirement Specification (SRS) Documentation   | /5   |
| Assumptions, Constraints, and Requirement Validation     | /5   |
| <b>TOTAL</b>                                             | /5   |

# 1. Scenario Understanding & Problem Analysis

## Business Domain

Construction Project Management

## Problem Summary

Construction companies currently manage multiple projects using spreadsheets and manual reporting. This leads to delayed material procurement, poor communication between site and office, and no real-time visibility into progress — resulting in cost overruns and missed deadlines.

## System Objectives

- Centralize project planning, resource management, and progress monitoring into a single platform
- Eliminate manual/spreadsheet-based reporting delays
- Enable real-time communication between site engineers, project managers, procurement, and clients
- Reduce cost overruns caused by late material ordering
- Provide transparent, verified progress visibility to clients tied to payment milestones

## Stakeholders

| Stakeholder         | Business Context                                                                                                                                         |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Site Engineer       | Works on-site, currently reports progress manually (paper/spreadsheet); needs a fast way to log data and flag material shortages                         |
| Project Manager     | Oversees multiple projects from the office; currently has no consolidated real-time view, relies on delayed manual reports                               |
| Procurement Team    | Currently reacts to requisitions late because they arrive via informal channels (calls, emails); needs a structured, trackable request-to-order pipeline |
| Client              | Currently receives progress updates infrequently and manually; wants transparent, verifiable milestone tracking before releasing payments                |
| Company Admin/Owner | Needs a portfolio-level view across all active projects to spot risk early                                                                               |
| Suppliers/Vendors   | External parties who fulfill purchase orders; interact indirectly through procurement's PO tracking                                                      |

## Existing Challenges

- No single source of truth for project status
- Manual reporting causes 1–2 day (or longer) lag between actual site events and management awareness
- Material shortages are discovered late, causing work stoppages
- Clients have low visibility, leading to payment disputes and trust issues
- No structured approval trail for requisitions or milestones

## 2. Functional Requirements (12)

| ID    | Requirement                                                                                                       |
|-------|-------------------------------------------------------------------------------------------------------------------|
| FR-1  | The system shall allow Site Engineers to log daily progress updates with photos and notes via a mobile app        |
| FR-2  | The system shall allow Site Engineers to update task completion status against the project schedule               |
| FR-3  | The system shall allow Site Engineers to raise material requisition requests from the field                       |
| FR-4  | The system shall allow Project Managers to view a consolidated dashboard of progress across all assigned projects |
| FR-5  | The system shall allow Project Managers to approve or reject material requisitions                                |
| FR-6  | The system shall allow Project Managers to view and adjust project schedules based on actual progress             |
| FR-7  | The system shall allow Project Managers to approve/verify site-reported milestones                                |
| FR-8  | The system shall allow Procurement Officers to convert approved requisitions into purchase orders                 |
| FR-9  | The system shall allow Procurement Officers to track and update purchase order delivery status                    |
| FR-10 | The system shall allow Clients to view real-time, verified milestone progress via a client portal                 |
| FR-11 | The system shall allow Clients to approve payments tied to verified milestones                                    |
| FR-12 | The system shall send notifications to relevant users when an action is pending (approval, delayed PO, etc.)      |

## Non-Functional Requirements (10)

| ID     | Category         | Requirement                                                                                                                                              |
|--------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-1  | Performance      | Progress updates submitted from the mobile app shall reflect on the PM dashboard within 5 minutes                                                        |
| NFR-2  | Usability        | The mobile interface shall be usable in low-connectivity site environments with minimal data entry steps                                                 |
| NFR-3  | Security         | All user data shall be protected via role-based access control (Site Engineer, PM, Procurement, Client, Admin)                                           |
| NFR-4  | Security         | Client payment approvals shall require authenticated login and be logged with timestamps                                                                 |
| NFR-5  | Reliability      | The system shall maintain 99.5% uptime during business hours                                                                                             |
| NFR-6  | Scalability      | The system shall support multiple concurrent projects and a growing number of users without performance degradation                                      |
| NFR-7  | Maintainability  | The system shall be built with modular components to allow independent updates to mobile, web dashboard, and portal modules                              |
| NFR-8  | Availability     | Core progress-logging and requisition features shall remain available even during partial network interruptions (offline caching with sync-on-reconnect) |
| NFR-9  | Interoperability | The system shall support data export (CSV/PDF) for reporting and audit purposes                                                                          |
| NFR-10 | Compliance       | The system shall retain audit trails of all approvals (requisitions, milestones, payments) for a minimum of 3 years                                      |

### **3. Actors & Use Cases**

#### **Primary Actors**

- Site Engineer
- Project Manager
- Procurement Officer
- Client

#### **Secondary Actors**

- Company Admin (oversight, not direct daily use)
- Supplier/Vendor (external, receives POs — not a direct system user in this scope)

#### **Use Cases per Actor**

##### **Site Engineer:**

- Log Daily Progress
- Update Task Status
- Raise Material Requisition
- View Assigned Schedule

##### **Project Manager:**

- View Consolidated Dashboard
- Approve/Reject Requisition
- Adjust Project Schedule
- Approve Milestone
- View Portfolio Summary (multi-project)

##### **Procurement Officer:**

- Convert Requisition to Purchase Order
- Track PO Delivery Status
- Update PO Status

##### **Client:**

- View Milestone Progress
- Approve Payment

##### **Company Admin (secondary):**

- View Cross-Project Summary Report

## Use Case Diagram:



Structure: 4 actors (Site Engineer, Project Manager, Procurement Officer, Client) connected via association lines to their respective use cases, all contained within a system boundary box labeled "Smart Construction Project Management System".

## **4. Software Requirement Specification (SRS)**

### **4.1 Introduction**

This SRS defines the requirements for the Smart Construction Project Management System, a centralized platform designed to replace manual, spreadsheet-based project tracking with real-time digital workflows connecting site engineers, project managers, procurement, and clients.

### **4.2 Problem Description**

Construction firms managing multiple simultaneous projects face delays and cost overruns due to fragmented, manual communication — particularly around material procurement and progress reporting. This system addresses that gap through role-specific interfaces feeding into one shared data source.

### **4.3 Scope of the System**

The system covers: mobile progress/requisition logging (Site Engineer), web-based schedule and approval management (Project Manager), purchase order tracking (Procurement), and a client-facing milestone/payment portal. It does not cover accounting/ERP integration, HR management, or direct supplier-side ordering systems (out of scope for this phase).

### **4.4 Functional Requirements**

See Section 2 (FR-1 to FR-12) above.

### **4.5 Non-Functional Requirements**

See Section 2 (NFR-1 to NFR-10) above.

### **4.6 Use Case Descriptions**

#### **UC-1: Log Daily Progress**

- Actor: Site Engineer
- Precondition: User is authenticated on the mobile app
- Main Flow: Engineer opens app → selects project/task → enters progress notes and photo → submits
- Postcondition: Progress entry stored, visible on PM dashboard

#### **UC-2: Approve Material Requisition**

- Actor: Project Manager
- Precondition: A requisition exists in "Pending" status
- Main Flow: PM opens requisition queue → reviews request details → approves or rejects with comment
- Postcondition: Status updated; Site Engineer and Procurement notified

#### **UC-3: Approve Payment**

- Actor: Client
- Precondition: Milestone status is "Verified"
- Main Flow: Client opens portal → reviews milestone → approves payment
- Postcondition: Payment status set to "Approved," logged with timestamp



## 4.7 Data Requirements

- User data: name, role, contact, project assignment
- Progress logs: timestamp, task ID, status, photo reference, notes
- Requisitions: material, quantity, urgency, status, linked project/task
- Purchase orders: PO number, linked requisition ID, delivery status, supplier reference
- Milestones: linked tasks, verification status, associated payment amount
- Notifications: recipient, trigger event, timestamp, read/unread status

## 4.8 External Interface Requirements

- Mobile App UI — for Site Engineers (Android/iOS)
- Web Dashboard UI — for Project Managers (browser-based)
- Client Portal UI — for Clients (browser-based, restricted view)
- Procurement Queue UI — for Procurement Officers (browser-based)
- Notification Interface — push notifications (mobile) and email/in-app alerts (web)
- No direct third-party hardware interfaces required in this phase

## 4.9 System Features

- Role-based dashboards per user type
- Real-time data sync across mobile and web
- Approval workflows with audit trail
- Milestone-linked payment gating (client can't pay for unverified work)
- Notification engine for pending actions

## 5. Assumptions and Constraints

### Assumptions

- All users have access to a smartphone (Site Engineers) or computer/browser (PM, Procurement, Client)
- Construction sites have at least intermittent internet connectivity for sync
- Project schedules are initially set up by the Project Manager before Site Engineers begin logging progress
- Payment amounts per milestone are predefined in the project contract, not calculated by the system

### Constraints

| Type        | Constraint                                                                                                                        |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Technical   | Mobile app must function with limited/offline connectivity (site environments)                                                    |
| Operational | Site Engineers may have limited technical literacy — UI must be simple and minimal-input                                          |
| Economic    | Solution must be cost-effective to deploy across multiple simultaneous project sites                                              |
| Legal       | Payment approval logs must be retained for audit/compliance purposes (see NFR-10)                                                 |
| Time        | Initial rollout (Sprint 1 scope) is limited to core field-to-office data flow; full feature set is phased across multiple sprints |

## **6. Requirement Validation**

### **Completeness Check**

- All 6 stakeholders have at least one functional requirement addressing their core need
- All 4 primary actors have associated use cases
- Every functional requirement maps to at least one user story from the Sprint Planning document

### **Consistency & Ambiguity Check**

Reviewed for conflicting requirements (e.g., no requirement contradicts another) — none found. FR-7 ("approve milestones") and FR-11 ("approve payment") were checked to ensure they are sequential, not overlapping — Project Manager verifies work; Client approves payment based on that verification, avoiding duplicate approval authority.

### **Redundancy Check**

No duplicate requirements identified across the 12 FRs.

### **Missing Requirements Identified & Added**

Initially, notification handling was not explicit — added as FR-12 and NFR-8 (offline handling) after review, since field connectivity issues would otherwise block core workflows.

### **Suggested Improvements**

- Add a future requirement for multi-language support, since site engineers across regions may not all be fluent in the same language
- Consider adding a dispute/rejection reason log for payment approvals, to support conflict resolution between client and contractor
- Add role-based reporting exports (e.g., PM-only cost reports) as a future enhancement beyond current scope.